

Ripudaman Singh

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Summary — Software Developer focused on building high-velocity backend systems, deploying cloud-native infrastructure, and optimizing pipeline metrics. Experienced in navigating fast-growing startup ambiguity to optimize product reliability and streamline developer workflows.

Skills

Infrastructure SST, AWS, Docker, Kubernetes, GitHub Actions
Observability Grafana LGTM Stack (Alloy, Mimir, Loki, Tempo), Langfuse, Posthog, Sentry, Mixpanel
Languages TypeScript, Python, C/C++, SQL, Go
Frameworks React, Vue, Flask, ExpressJS, FastAPI, Hono, SQLAlchemy, Drizzle

Experience

- trophi.ai** Jan 2026 – Jun 2026
Software Developer (Full Time)
- Deployed scalable, cloud-native infrastructure using SST and AWS, driving initiatives to optimize platform reliability and minimize downtime.
 - Designed and delivered an end-to-end, real-time telemetry coaching pipeline, integrating RAGFlow and Langfuse for deep context retrieval, LLM observability, and automated evaluation.
 - Instrumented and deployed the Grafana LGTM stack to centralize infrastructure metrics, streamlining alerting and accelerating developer debugging velocity.
- Visual and Analytic Computing Lab, Memorial University** May 2023 – Dec 2024
Research Assistant (Full Time) May 2024 — Dec 2024
- Designed and conducted a user study for 30 participants
 - Developed data clean-up, aggregation and analysis solution for the user study using Pandas
- Software Developer (CO-OP, Full Time)* May 2023 — Dec 2023
- Developed a flight path visualization XR application in Unreal Engine 5 for Microsoft HoloLens 2
 - Implemented interfacing using UMG and the ability to manipulate flight paths through hand tracking
- Data and Image Analysis Group, Memorial University** Jan 2023 – Apr 2023
Data Quality Assurance Analyst
- Developed an SSH manager utility using Python to help setup systems seamlessly
 - Contributed to a scheduler for a HPC payload processing system
- Department of Science, Memorial University** Sep 2022 – Dec 2022
Teaching Assistant (Computer Science)
- Assisted first-year students with core computer science concepts such as DFAs, Turing Machines and Regular Expressions
 - Graded course work, conducted lab sessions and provided feedback for 40+ students
- Student Residences, Memorial University** Jan 2022 – Apr 2024
Resident Assistant (Full Time)
- Oversaw and guided 100 students per semester
 - Planned and conducted inclusive and welcoming events to help ease the transition to university life

Projects

- MediSync AI** | ExpressJS, Flask, MongoDB, OpenAI API, React, RAG May 2024
- Won Best Overall at the DO.IT Hackathon as an aide to healthcare professionals, facilitating them to perform CRUD operations using natural language and without the need for technical experience
- Honours Thesis Project** | Python, matplotlib, opencv, 3D Slicer, tobii Dec 2024
- Developed a framework for collecting and visualizing eye-tracking data for medical image analysis
 - Automated loading of DICOM images and ROIs in 3D Slicer and visualizing the spatial data using heatmaps
- Bubblegum Bloodbath (C++ Game)** | C++, SFML, ImGui Dec 2024
- Developed a C++ game from scratch using SFML as part of a group project
 - Won Best Overall in the batch
- Car Rental Full Stack** | TypeScript, React, Chakra UI, Node.js, Express, MySQL Dec 2024
- A full-stack car rental application with user authentication, vehicle management, and booking system

Education

Memorial University of Newfoundland Jan 2021 – Feb 2025
Bachelor of Science – Computer Science, Honours; GPA: 4.0